AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION 2010

General Mathematics Paper I

Time allowed: 40 minutes Marks 30

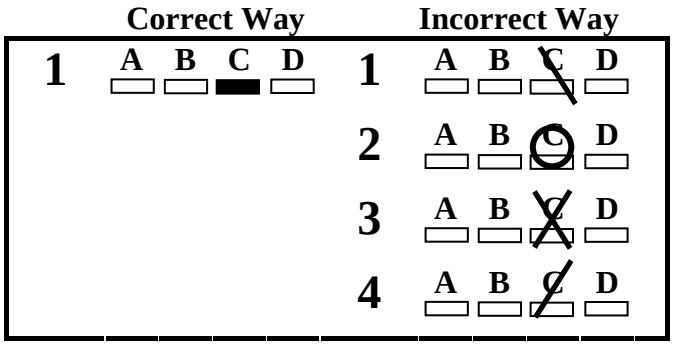
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

7. You may use a simple calculator if you wish.

1. The fraction $\frac{3}{5}$ is equivalent to

   A. 75%
   B. 60%
   C. 20%
   D. 5%

2. The ratio between the ages of Nazia and Shazia is 2:3. If Nazia is 10 years old, the age of Shazia is

   A. 12 years.
   B. 13 years.
   C. 15 years.
   D. 18 years.

3. In percentage, 0.35 is equal to

   A. 3.5%
   B. 35%
   C. 70%
   D. 75%

4. Sameer has Rs.1,00,000 in his bank account. The amount of zakat to be deducted from his account on 1st Ramazan will be [Note: Zakat is 2.5% of the total annual saving]

   A. Rs. 1,000
   B. Rs. 2,500
   C. Rs. 10,000
   D. Rs. 25,000

5. What will be the amount of Usher due on a cotton yield of Rs.265,000?
   [Note: Usher is 10% of the yield]

   A. Rs. 26.50
   B. Rs. 265.00
   C. Rs. 2,650
   D. Rs. 26,500



6. *A* and *B* are two business partners. They made a profit of Rs.60,000. If *A* has twice as many share as *B*, then the profit of *A* is equal to

   A. Rs. 20,000
   B. Rs. 30,000
   C. Rs. 40,000
   D. Rs. 50,000

7. Marked price of a book is Rs.120. If a shopkeeper gives a discount of 10% on marked price, then the sale price of the book is



A. Rs. 100
B. Rs. 108
C. Rs. 110
D. Rs. 112

8. Saleem sold a telephone set for Rs.1250 having a cost price of Rs.1500. His loss percentage on this sale is

A. 12.67%
B. 14.67%
C. 16.67%
D. 18.67%

9. Mr. Shamsi has Euro 50. The amount he will get in Pakistani Rupees at an exchange rate of Rs.120 per Euro is

A. 6,500
B. 6,000
C. 5,500
D. 5,000

10. ATM stands for



A. Automatic teller machine
B. Accessible transaction machine
C. Automatic transaction machine
D. Accessible transfer machine

11. If Rs.19, 000 is exchanged for $225, then 1 dollar is equal to



A. Rs. 80.44
B. Rs. 81.44
C. Rs. 84.44
D. Rs. 87.44

12. Which of the following is not an appropriate option to complete the statement? “Insurance is a sort of risk management to

A. save the job”.
B. hedge the loss”.
C. transfer risk of loss to another party”.
D. provide coverage against mishap”.

13. If 15% sales tax is levied on a product costing Rs.200, then the consumer price is

A. Rs. 200
B. Rs. 215
C. Rs. 230
D. Rs. 245

14. The price of a tennis ball with 5% sales tax is Rs. 84. If tax is withdrawn on sports items, then the price of the ball without tax is

A. Rs. 79.0
B. Rs. 80.0
C. Rs. 82.2
D. Rs. 88.2

15. The sales tax charged on a shirt at the rate of 16%, is Rs.160. The price of the shirt in Rupees is

A. 800
B. 875
C. 925
D. 1000

16. In an electric bill, the previous meter reading was 923452 and current reading is 924232. If energy charges are Rs.7 per unit excluding all other charges, then the total amount of energy charges in Rupees is

A. 4,780
B. 4,860
C. 5,280
D. 5,460

17. $\left(\frac{a^{-1}}{a}\right)^2$ is equal to

A. 1
B. a^4
C. $\frac{1}{a^2}$
D. $\frac{1}{a^4}$

18. $\log_{10} 10000$ is equal to

A. 5
B. 4
C. 3
D. 2

19. If $\log 8 = x$, then the value of $\log 2$ is

A. x^3
B. 3
C. $3x$
D. $\frac{x}{3}$

20. The geometric mean between 4 and 64 is

A. 8
B. 16
C. ± 8
D. ± 16

21. If a is the arithmetic mean of 6 and 10, then $a - 6$ is equal to

A. – 4
B. – 2
C. 2
D. 4

22. The 5th term of the geometric progression 3, 9, 27, 81 is

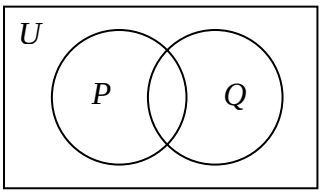
A. 90
B. 99
C. 243
D. 729

23. For two sets $A = \{0,1\}$ and $B = \{1,2\}$, which one of the following is a function from A to B?

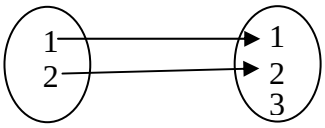
A. $\{(1,1)\}$
B. $\{(0,1),(0,2),(1,1),(1,2)\}$
C. $\{(0,1),(0,2),(1,2)\}$
D. $\{(0,2),(1,2)\}$

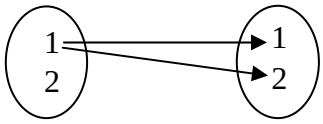
24. Consider the following Venn diagram. $P \cup (Q \cap P)$ is equal to

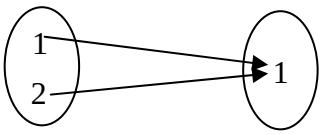
A. P
B. Q
C. $P - Q$
D. $Q - P$

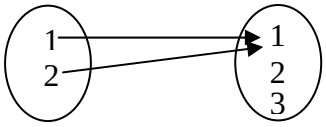


25. Which one of the following figures represents an injective function?

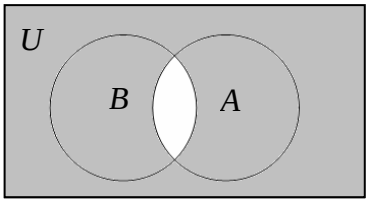
A. 

B. 

C. 

D. 

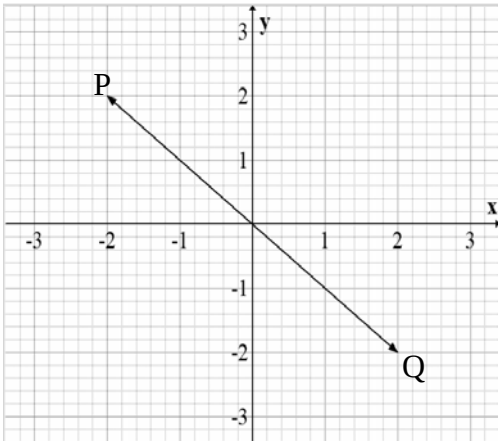
26.



The shaded region (grey) in above Venn diagram represents

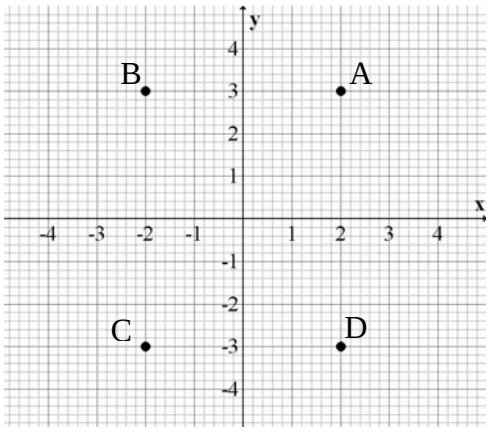
A. $A \cap B$
B. $A \cup B$
C. $(A \cup B)'$
D. $(A \cap B)'$

27. In the following graph, the equation of the line PQ is



A. $y = -x$
B. $y = x$
C. $y = x - 1$
D. $y = 1 - x$

28. In the following graph, which of the points has coordinates as (2, –3)?

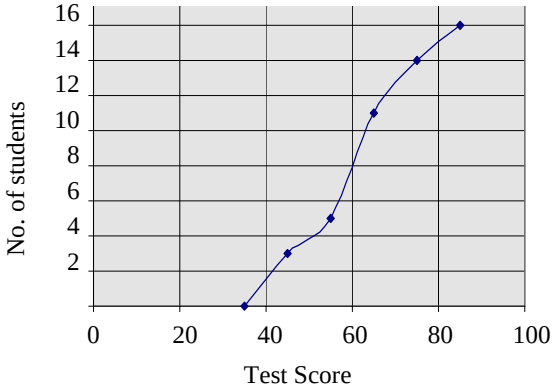


29. The median of four numbers is 6.5. If three of the numbers are 3, 5 and 9, then the 4th number is

A. 10
B. 8
C. 4
D. 3

30. The following cumulative frequency polygon shows the test scores of 16 students. The median of the test scores is

A. 8
B. 50
C. 60
D. 65



END OF PAPER

Please use this page for rough work